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Differentiation Practice Sheet

Nine Education IIT Academy | Class 11

SECTION A — Direct Formulae (Q 1–15)

Differentiate each function with respect to x.

1. $y = x^9$
2. $y = x^{(-5)}$
3. $y = \sqrt{x}$
4. $y = x^{(3/2)}$
5. $y = \sqrt[3]{x}$ *(i.e., $x^{(1/3)}$)*
6. $y = 1/x^4$
7. $y = e^x$
8. $y = \ln x$
9. $y = \sin x$
10. $y = \cos x$
11. $y = \tan x$
12. $y = \cot x$
13. $y = \sec x$
14. $y = \operatorname{cosec} x$
15. $y = 6x^5$

SECTION B — Sum & Difference (Q 16–40)

Differentiate each function with respect to x.

16. $y = x^3 + x^2$
17. $y = 3x^4 - 5x^2 + 7$
18. $y = x^3 - \sin x + 2$
19. $y = e^x + \ln x$
20. $y = \cos x + \tan x$
21. $y = x^2 + e^x - \ln x$
22. $y = 5x^3 - 3x^2 + 2x - 1$
23. $y = \sqrt{x} + 1/\sqrt{x}$
24. $y = x^2 - \cos x + \sin x$

25. $y = x^{-1} + x^{-2} + x^{-3}$
26. $y = 2e^x - 3 \ln x + 5$
27. $y = \sin x - \cos x + \tan x - \cot x$
28. $y = x^{5/2} - x^{3/2} + x^{1/2}$
29. $y = \sec x + \operatorname{cosec} x$
30. $y = 4x^3 - 2/x + 5$
31. $y = x^2 + 1/x^2 - 2$
32. $y = e^x + \sin x - \cos x + \ln x$
33. $y = x^{1/3} + x^{2/3} - x$
34. $y = 3 \sin x - 4 \cos x + 2 \tan x - \operatorname{cosec} x$
35. $y = x^4 - 2x^3 + 3x^2 - 4x + 5$

[Expand / Simplify first, then differentiate]

36. $y = (x^3 + 1) / x^2$
37. $y = (x^2 + 1)^2$
38. $y = (\sqrt{x} - 1/\sqrt{x})^2$
39. $y = (x + 1)(x - 1)(x^2 + 1)$
40. $y = 2x^{4/3} - 3x^{2/3} + 6x^{-1/3} - 5$

SECTION C — Product Rule (Q 41–60)

Use: d/dx (uv) = u'v + uv'

41. $y = x \cdot e^x$
42. $y = x^2 \sin x$
43. $y = x \ln x$
44. $y = x^3 \cos x$
45. $y = e^x \sin x$
46. $y = e^x \cos x$
47. $y = x \tan x$
48. $y = x^2 \ln x$
49. $y = \sin x \cdot \cos x$
50. $y = e^x \ln x$
51. $y = (x^2 + 1) \sin x$
52. $y = x \sec x$
53. $y = (x^3 - 2x) e^x$
54. $y = x^2 e^x$
55. $y = (1 + \tan x) e^x$
56. $y = (x^2 - 3x + 2)(x + 4)$

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57. $y = x \operatorname{cosec} x$
58. $y = \ln x \cdot \sin x$
59. $y = (x + \sin x) \cos x$
60. $y = (x^2 + x + 1)(x^2 - x + 1)$

SECTION D — Quotient Rule (Q 61–80)

Use: $d/dx\ (u/v) = (u'v - uv') / v^2$

61. $y = \sin x / x$
62. $y = e^x / x$
63. $y = \ln x / x$
64. $y = x / e^x$
65. $y = \sin x / \cos x$ *(Derive from scratch — do not quote the answer directly)*
66. $y = x^2 / \sin x$
67. $y = e^x / (x + 1)$
68. $y = (x^2 - 1) / (x^2 + 1)$
69. $y = \ln x / e^x$
70. $y = (x + 1) / (x - 1)$
71. $y = \cos x / e^x$
72. $y = (x^2 + x + 1) / (x^2 - x + 1)$
73. $y = \tan x / x$
74. $y = x / \ln x$
75. $y = (x^3 + 1) / (x + 1)$
76. $y = \sin x / (1 + \cos x)$
77. $y = (e^x - 1) / (e^x + 1)$
78. $y = (x^2 - x) / (x + 1)$
79. $y = \operatorname{cosec} x / x$
80. $y = (\sin x + \cos x) / (\sin x - \cos x)$

SECTION E

These problems require multiple rule applications. Show every step clearly.

81. $y = x \cdot e^x \cdot \sin x$
82. $y = x^2 \cdot \ln x \cdot \cos x$
83. $y = (x^2 + \sin x)(e^x - \cos x + \ln x)$
84. $y = (x + e^x)(x^2 - \ln x + \sin x)$

85. $y = (e^x \sin x) / (x^2 + \ln x)$
86. $y = (x \ln x + e^x) / (x + \sin x)$
87. $y = (x \sin x + \cos x) / (x e^x + \ln x)$
88. $y = x \cdot e^x \cdot \ln x \cdot \sin x$
89. $y = [(x^2 + 1) \sin x] / [(x + 1) e^x]$
90. $y = [(x e^x + \sin x)(\ln x - \cos x)] / (x^2 + 1)$

Answer Key

Section A (Q 1–15)

Q	Answer
1	$9x^8$
2	$-5x^{-6} = -5/x^6$
3	$1/(2\sqrt{x})$
4	$(3/2)\sqrt{x} = (3/2)x^{1/2}$
5	$(1/3)x^{-2/3} = 1/(3\sqrt[3]{x^2})$
6	$-4/x^5$
7	e^x
8	$1/x$
9	$\cos x$
10	$-\sin x$
11	$\sec^2 x$
12	$-\mathrm{cosec}^2 x$
13	$\sec x \tan x$
14	$-\mathrm{cosec}\, x \cot x$
15	$30x^4$

Section B (Q 16–40)

Q	Answer
16	$3x^2 + 2x$

Q	Answer
17	$12x^3 - 10x$
18	$3x^2 - \cos x$
19	e^x + 1/x
20	$-\sin x + \sec^2 x$
21	2x + e^x − 1/x
22	$15x^2 - 6x + 2$
23	1/(2√x) − 1/(2x^(3/2))
24	2x + sin x + cos x
25	$-1/x^2 - 2/x^3 - 3/x^4$
26	2e^x − 3/x
27	$\cos x + \sin x + \sec^2 x + \mathrm{cosec}^2 x$
28	(5/2)x^(3/2) − (3/2)x^(1/2) + 1/(2√x)
29	sec x tan x − cosec x cot x
30	$12x^2 + 2/x^2$
31	$2x - 2/x^3$
32	e^x + cos x + sin x + 1/x
33	(1/3)x^(−2/3) + (2/3)x^(−1/3) − 1
34	$3 \cos x + 4 \sin x + 2 \sec^2 x + \mathrm{cosec}\, x \cot x$
35	$4x^3 - 6x^2 + 6x - 4$
36	$1 - 2/x^3$ *(Simplify: y = x + x^(−2))*
37	$4x^3 + 4x$ *(Expand: $y = x^4 + 2x^2 + 1$)*
38	$1 - 1/x^2$ *(Expand: y = x − 2 + x^(−1))*
39	$4x^3$ *(Expand: $y = x^4 - 1$)*
40	(8/3)x^(1/3) − 2x^(−1/3) − 2x^(−4/3)

Section C (Q 41–60)

Q	Answer
41	e^x(1 + x)
42	$2x \sin x + x^2 \cos x$
43	ln x + 1
44	$3x^2 \cos x - x^3 \sin x$

Q	Answer
45	e^x(sin x + cos x)
46	e^x(cos x − sin x)
47	tan x + x sec^2x
48	x(2 ln x + 1)
49	$cos^2x - sin^2x$
50	e^x(ln x + 1/x)
51	2x sin x + (x^2 + 1) cos x
52	sec x(1 + x tan x)
53	e^x($x^3 + 3x^2 - 2x - 2$)
54	xe^x(x + 2)
55	e^x(1 + tan x + sec^2x)
56	$3x^2 + 2x - 10$
57	cosec x(1 − x cot x)
58	sin x / x + ln x · cos x
59	cos x + cos^2x − x sin x − sin^2x
60	$4x^3 + 2x$ *(Expand: $y = x^4 + x^2 + 1$)*

Section D (Q 61–80)

Q	Answer
61	(x cos x − sin x) / x^2
62	e^x(x − 1) / x^2
63	(1 − ln x) / x^2
64	(1 − x) / e^x
65	sec^2x
66	x(2 sin x − x cos x) / sin^2x
67	xe^x / $(x + 1)^2$
68	4x / $(x^2 + 1)^2$
69	(1 − x ln x) / (x e^x)
70	−2 / $(x - 1)^2$
71	−(sin x + cos x) / e^x
72	2(1 − x^2) / $(x^2 - x + 1)^2$

Q	Answer
73	$(x \sec^2 x - \tan x) / x^2$
74	$(\ln x - 1) / (\ln x)^2$
75	$(2x^3 + 3x^2 - 1) / (x + 1)^2$
76	$1 / (1 + \cos x)$
77	$2e^x / (e^x + 1)^2$
78	$(x^2 + 2x - 1) / (x + 1)^2$
79	$-\operatorname{cosec} x(1 + x \cot x) / x^2$
80	$-2 / (\sin x - \cos x)^2$

Section E (Q 81–90)

Q81. $dy/dx = e^x [\sin x + x(\sin x + \cos x)]$

Q82. $dy/dx = x [(2 \ln x + 1) \cos x - x \ln x \cdot \sin x]$

Q83. $dy/dx = (2x + \cos x)(e^x - \cos x + \ln x) + (x^2 + \sin x)(e^x + \sin x + 1/x)$

Q84. **$dy/dx = (1 + e^x)(x^2 - \ln x + \sin x) + (x + e^x)(2x - 1/x + \cos x)$**

Q85. $dy/dx = e^x [(\sin x + \cos x)(x^2 + \ln x) - \sin x(2x + 1/x)] / (x^2 + \ln x)^2$

Q86. $dy/dx = [(\ln x + 1 + e^x)(x + \sin x) - (x \ln x + e^x)(1 + \cos x)] / (x + \sin x)^2$

Q87. **$dy/dx = [x \cos x \cdot (x e^x + \ln x) - (x \sin x + \cos x)(e^x(1+x) + 1/x)] / (x e^x + \ln x)^2$**

Q88. $dy/dx = e^x [\sin x (1 + \ln x (1 + x)) + x \ln x \cdot \cos x]$

Q89.$**$ $dy/dx = [-(x^3 - x + 2) \sin x + (x + 1)(x^2 + 1) \cos x] / [(x + 1)^2 e^x]$$**$

Q90.

> $dy/dx = \{ [e^x(1+x) + \cos x](\ln x - \cos x) + (x e^x + \sin x)(1/x + \sin x) \}(x^2 + 1) - 2x(x e^x + \sin x)(\ln x - \cos x)$
>
> ---
>
> ——— $(x^2 + 1)^2$

End of Sheet